EXPERIMENT 09

SIMULATION OF TELEMETRY DATA INGESTION TO TIME-SERIES DATABASE AND VISUALIZATION USING GRAFANA, POWER BI AND TABLEAU

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1. AIM OF THE EXPERIMENT

To design, simulate, and analyze high-frequency multi-sensor Internet of Everything (IoE) telemetry data ingestion into an optimized Time-Series Database (DuckDB/TSDB), and evaluate multi-panel real-time operational surveillance, ISO 10816 vibration severity standards, cross-sensor correlations, thermodynamic phase space trajectories, and hourly aggregation envelopes across industrial analytics dashboards.

2. OBJECTIVES

Implement a high-throughput time-series ingestion engine simulating a 48-hour continuous industrial carbon capture monitoring window (8,640 samples).

Acquire multi-variable telemetry streams: tri-node CO_2 concentration (IOE-CO2-01 to 03), motor vibration velocity (mm/s RMS), thermodynamic temperature (°C), barometric pressure (hPa), and relative humidity (%).

- Establish multi-tier threshold alerting: CO_2 baseline warning (1,000 ppm), critical surge alarm (1,800 ppm), and ISO 10816 Zone D critical vibration threshold (> 7.1 mm/s RMS).
- Compute bivariate Pearson cross-correlation coefficients across all physical sensor modalities.
- Map multi-state thermodynamic phase boundaries (Normal, Warning, Critical) and benchmark 1-hour downsampled aggregations with 95th-percentile (p95) peak tracking envelopes.

3. MATHEMATICAL MODEL AND FORMULAE

1. Ingestion Throughput (R_{ingest}):	$R_{ingest} = N_{total} / T_{elapsed}$ (records/second, where N = 8,640 records)
2. Vibration Severity RMS Velocity:	$v_{RMS} = \sqrt{[\ (1 / K) \times \sum_{k=1}^{K} v_k^2\ ]}$ (evaluated against ISO 10816-3 criteria)
3. Pearson Cross-Correlation (r_{xy}):	$r_{xy} = \sum(x_i - \bar{x})(y_i - \bar{y}) / [\ \sqrt{\sum(x_i - \bar{x})^2} \times \sqrt{\sum(y_i - \bar{y})^2}\ ]$
4. Rolling 1-Hour Downsampling:	$CO2_{1\text{-hr}} = (1 / W) \times \sum_{j=1} W\ CO2_j$ (W = 180 samples per 1-hour interval)
5. Multi-Sensor Operational Rules:	Normal: $CO_2 \leq 1{,}000$ ppm & $v_{RMS} < 1.8$ mm/s; Critical: $CO_2 > 1{,}800$ ppm or $v_{RMS} > 7.1$ mm/s

4. SOFTWARE & TOOLS REQUIRED

Python 3.10+ (Libraries: pandas, numpy, duckdb, matplotlib, seaborn, scipy), Time-Series Storage Engine (DuckDB / InfluxDB v2.x), Grafana Open Source / Cloud, Microsoft Power BI Desktop, and Tableau Public.

5. EXPERIMENTAL PROCEDURE

1. Initialize the time-series storage workspace and define a columnar table schema in DuckDB capturing timestamp, device_id, co2_ppm, temp_c, humidity_pct, pressure_hpa, and vibration_rms.
2. Configure a continuous synthetic telemetry engine generating 48 hours of time-stamped sensor data at a 20-second sampling cadence, totaling 8,640 observations across nodes IOE-CO2-01, IOE-CO2-02, and IOE-CO2-03.
3. Inject realistic diurnal periodic trends, thermodynamic expansion profiles, scheduled industrial gas release spikes (> 3,000 ppm), and motor bearing vibration excursions (11.30 mm/s RMS).

4. Execute analytical SQL queries in DuckDB to compute summary KPIs: total sample count, global mean CO_2, peak temperature, peak vibration velocity, and excursion alarm counts.
5. Construct Panel 1 (Real-Time Tri-Node CO_2 Telemetry & Surge Thresholds) with warning (1,000 ppm) and critical alarm (1,800 ppm) boundaries.
6. Construct Panel 2 (ISO 10816 Motor Vibration Severity) classifying vibration into Zone A (< 1.8), Zone B (1.8–4.5), Zone C (4.5–7.1), and Zone D (> 7.1 mm/s RMS).
7. Compute and render Panel 3 (Multi-Sensor Cross-Correlation Matrix) across all five sensor pairs.
8. Plot Panel 4 (Thermodynamic State Space) mapping Pressure vs. Temperature categorized into Normal, Warning, and Critical clusters.
9. Generate Panel 5 (DuckDB 1-Hour Downsampled Aggregation) computing hourly averages, peak envelopes, and the 95th-percentile (p95) tracking contour.

6. INPUT PARAMETER TABLE

S.NO.	INPUT PARAMETER	CONFIGURATION SYMBOL	APPLIED VALUE	UNIT
1	Time-Series Database Storage	Database Engine	DuckDB (In-Memory TSDB)	—
2	Ingestion Surveillance Window	T_{window}	48.0 (11-01 00:00 to 1103 00:00)	Hours
3	Total Ingested Records	N_{total}	8,640	Samples
4	Monitored Sensor Nodes	Node IDs	IOE-CO2-01, IOE-CO2-02, IOE-CO2-03	—
5	CO_2 Baseline Warning Limit	$CO_{2,warn}$	1,000	ppm
6	CO_2 Critical Surge Alarm	$CO_{2,crit}$	1,800	ppm
7	Vibration Severity Zone D Boundary	v_{crit}	7.1 (ISO 10816 Limit)	mm/s RMS
8	Permissible Operating Temperature	T_{limit}	85.0	°C
9	Downsampling Aggregate Window	$W_{downsample}$	1.0	Hour

7. EXPERIMENTAL OUTPUT IMAGES



Figure 1: IoE Carbon Capture & Multi-Sensor Telemetry Dashboard displaying 48h Ingested Window KPIs, Surge Monitoring, ISO 10816 Vibration Severity, Cross-Correlation Matrix, Thermodynamic State Space, and DuckDB 1-Hour Downsampled Aggregation.

8. TABULAR RESULT & OBSERVATION RECORD

PANEL	MEASURED METRIC / PARAMETER	EXPERIMENTAL VALUE	REFERENCE / THRESHOLD	STATUS / ZONE
Header KPI	Total Telemetry Ingestion	**8,640 Samples**	48h Time-Series Window	COMPLETE
Header KPI	Mean CO_2 Concentration	**930.8 ppm**	Peak Surge: 3,006 ppm	MONITORED
Header KPI	Peak Operational Temperature	**64.8 °C**	Operating Limit: 85.0 °C	NORMAL (< 85°C)
Header KPI	Peak Vibration Severity Velocity	**11.30 mm/s**	ISO 10816 Limit: 7.10 mm/s	**ZONE D (CRITICAL)**
Header KPI	Total Critical Alarm Excursions	**358 Alarms**	Surge / Bearing Excursions	FLAGGED & LOGGED
Panel 1	CO_2 Diurnal Baseline (Troughs)	450 − 550 ppm	Warning Line: 1,000 ppm	NORMAL OPERATION
Panel 1	CO_2 Major Surge Spikes (08:00 & 07:00)	**2,850 − 3,006 ppm**	Alarm Line: 1,800 ppm	**CRITICAL EXCURSION**
PANEL	**MEASURED METRIC / PARAMETER**	**EXPERIMENTAL VALUE**	**REFERENCE / THRESHOLD**	**STATUS / ZONE**
Panel 2	Baseline Motor Vibration Velocity	1.2 − 1.8 mm/s	ISO Zone A: < 1.8 mm/s	ZONE A (GOOD)
Panel 2	Transient Bearing Fault Spike (20:00)	**11.30 mm/s**	ISO Zone D: > 7.1 mm/s	**ZONE D (CRITICAL)**
Panel 3	Cross-Correlation: CO_2 vs. Temperature	**+0.88**	Pearson Coefficient [-1 to +1]	STRONG POSITIVE
Panel 3	Cross-Correlation: CO_2 vs. Humidity	**−0.59**	Pearson Coefficient [-1 to +1]	MODERATE NEGATIVE
Panel 3	Cross-Correlation: Pressure vs. Vibration	**−0.05**	Pearson Coefficient [-1 to +1]	UNCORRELATED
Panel 4	Thermodynamic State Boundary	1008 − 1018 hPa | 20 − 65 °C	State Classification Mapping	3 DISTINCT CLUSTERS
Panel 5	DuckDB Downsampled Peak Envelope	**~2,450 ppm**	p95 vs. Hourly Average	HIGH CONVERGENCE

9. RESULT ANALYSIS & INTERPRETATION

Time-Series Ingestion & Storage Scalability: The DuckDB columnar storage engine ingested all 8,640 records across 48 continuous monitoring hours with sub-millisecond query retrieval latency, verifying seamless ingestion of multi-sensor industrial data streams.

- **Tri-Node CO_2 Dynamics & Surge Detection:** Sensors IOE-CO2-01, 02, and 03 recorded diurnal rhythmic cycles between 450 ppm and 1,450 ppm. Two major industrial process excursions on 11-01 and 11-02 breached both the 1,000 ppm warning threshold and the 1,800 ppm critical limit, peaking at 3,006 ppm and triggering 358 recorded alarms.
- **Vibration Severity Conformance (ISO 10816):** While continuous baseline motor operations remained securely within Zone A (< 1.8 mm/s RMS), an isolated mechanical excursion reached 11.30 mm/s RMS on 11-01 (~20:00), crossing Zone C (warning) into Zone D (critical), indicating an imminent bearing fault.

- **Thermodynamic & Cross-Correlation Insights:** A strong positive correlation ($r = +0.88$) was established between CO_2 concentration and temperature, reflecting thermal expansion and release kinetics, while relative humidity demonstrated an inverse relationship ($r = -0.59$). The thermodynamic state-space plot delineated normal operating bands from elevated warning and critical clusters.
- **Downsampling Fidelity:** The DuckDB 1-hour downsampling pipeline tracked global trends while the 95th-percentile (p95) envelope preserved peak surge visibility (~2,450 ppm), ensuring data reduction without loss of critical safety transients.

10. CONCLUSION

The simulation of telemetry data ingestion to an optimized time-series database and multi-panel visualization was successfully implemented and evaluated. The system processed 8,640 multi-sensor records across a 48-hour continuous window, accurately mapped ISO 10816 motor vibration severity, identified CO_2 surge excursions exceeding 3,000 ppm, and validated cross-sensor correlation kinetics with high statistical fidelity.